

Data Plots: Hydrolysis of Nitrosamines

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This exploration provides discussion for one of the cases presented in class meeting #14. In the class handout you were asked make a Hammett plot and come to a conclusion about the reaction mechanism. This document explores that data and ways to interpret it by plotting and fitting the data to mathematical models.¹

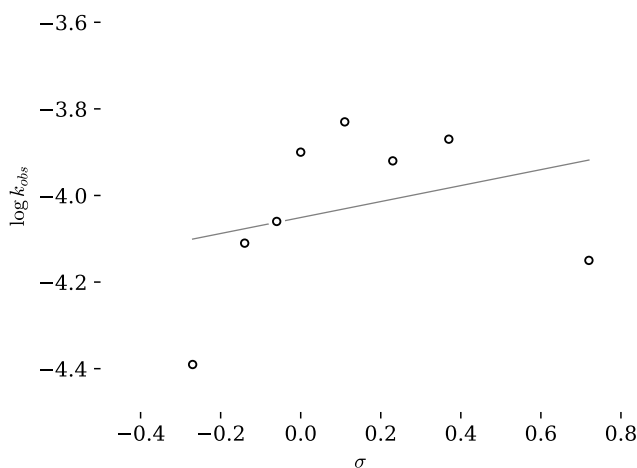
Plots: Curved Hammett Plots

This data comes from an investigation into the mechanism of hydrolysis of phenylimines of salicylaldehydes² This is a fine subject for a student exploration. Someday I will write one up. There are many rabbit holes to go down here.

This document will simply present the plots that were discussed in class. The plots are generated using a *Python* notebook available via Google Colab.³

The Basic Plot

Below is a Hammett plot of $\log k_{obs}$ vs σ . As you can see, a simple line plot may not be useful here.



This document was produced using the $\text{\LaTeX}$ typesetting language with the Tufte-handout document class. Chemical diagrams were created in *ChemDoodle* Plots were typeset using *Matplotlib* and *Python* tools. Diagrams were created and edited using *Affinity Designer*.

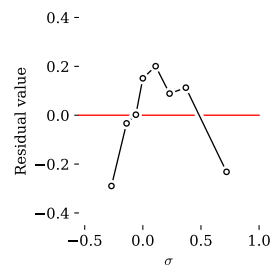
¹ Read chapter 8.3 and 8.4.

² "Kinetics of hydrolysis of substituted salicylideneanilines." J. Hoffmann, J. Klicnar, V. Štěrbá, M. Večeřa, *Coll. Czech. Chem. Comm.*, 1970, 35, 1387–1398. <http://doi.org/10.1135/cccc19701387>

³ All plots in this exploration were generated using the interactive *Python* notebook available via Google colab at https://colab.research.google.com/github/blinkletter/4410PythonNotebooks/blob/main/Class.16/16.Class_Handout_Problems_Imines.ipynb.

Figure 1: plot of $\log k_{obs}$ vs. σ . Do you see a bend in the plot? If we fit to a straight line we see that $\rho = 0.18 \pm 0.45$, $R^2 = 0.10$.

Figure 2: A plot of the residuals vs. σ for the line fit in figure 1. We observe a systematic variation from the model, not random noise.



Two Separate Plots

Below is a Hammett plot of $\log k_{obs}$ vs σ . As you can see, I divided the points into two groups and performed a line fit on each. This gave two separate Hammett equations.

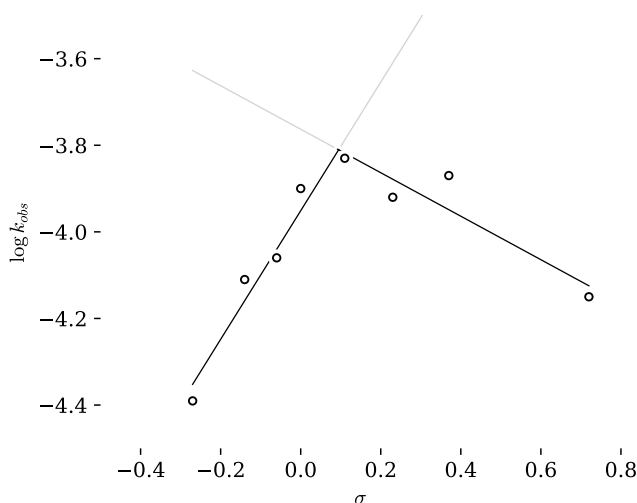


Figure 3: plot of $\log k_{obs}$ vs. σ . I bet you see the bend in the plot now. Of course, I am directing your eye by drawing the two lines.

$$y = (1.49 \pm 0.19)x + (-3.95 \pm 0.03), R^2 = 0.954$$

$$y = (-0.50 \pm 0.14)x + (-3.76 \pm 0.06), R^2 = 0.858$$

That's about it. I have no mathematical model for fitting the curve fit of two lines where the lesser controls the rate. I am sure that I could come up with a rate law that could be fitted. I would have to use the Eyring equation and change the free energy with some function according to σ for all the rate constants involved. I haven't gotten around to that yet and it does seem like a lot of work to implement. In the end, estimating the two slopes is pretty easy as long as you have enough data.

Interpretation

So how do you interpret a Hammett plot where the line is "concave down" and switches from a slope of about $\rho = 1.5$ to a slope of about $\rho = -0.5$? Answer the questions in the class meeting handout and you will find your way.

Exploration Tips

This exploration has just begun. We have plotted the Hammett plot using data in the paper and found it to be consistent with the figure in the paper. My own line fits used the top data point twice (five points in left line, four points in right line). Was that justified? Feel free to criticize my decision.

Exploration Questions

There is a lot to explore here.

1. **RATE DETERMINING STEP:** The authors lay out a detailed mechanism and provide an argument for how the rds changes. There is an analysis of the ρ value for each of the four steps in the proposed mechanism. The authors are not consistent with their symbols and that could be discussed also. Its always fun to pick at small details.
2. **pH-RATE PROFILES:** The authors report a pH-rate profile. It is different than typical profiles for imine hydrolysis, likely because of the phenol group of the salicylaldehyde component. This would be fun to explore.
3. **BILL JENCKS:** The authors reference the work of William Jencks, a giant of physical-organic chemistry, who investigated many mechanisms for imines and related groups. You could explore some of this work to highlight how this paper proposes a slightly different mechanistic analysis.
4. **BASE-CATALYZED MECHANISM:** There is an analysis of the Hammett plot for the base-catalyzed mechanism also. This analysis could be explored as well.
5. **TAFT PARAMETERS:** The authors use Taft's substituent parameters (described as σ^0 in this paper, but now usually denoted as σ^* .) Why did they use Taft parameters rather than Hammett parameters for the LFER investigating the effect of substituents on the K_a value of the phenol group in the salicylideneanilines? What did the plot reveal? Replot it with the modern sets of σ^* values (nothing will change – at least nothing should.)
6. **TOO MUCH WORK:** I had suggested that perhaps I could come up with a mathematical model to fit the Hammett plot. Using the ρ values proposed for all the steps (and perhaps trying out various values for the k_{-2} step) there is an opportunity. But will it work out?

As you can see there was lots more that could have been done. I would not be able to explore it all in the 10-page body of work that I am expecting in a student exploration report. Pick one, two or three things and dig in to them. If your paper is short doesn't have much to go on you can choose another paper or explore one of the references to you paper.